

Assessment of Candidate Technology ETSI: DECT-2020 New Radio

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Abstract—A number of IMT-2020 candidate technologies including the 3GPP have been submitted to the ITU-R. All candidate technologies are expected to meet the minimum performance requirements set by ITU-R for getting recommended as IMT-2020. In this paper, based on the submissions for Digital Enhanced Cordless Telecommunication (DECT-2020 NR) technology, we have evaluated the performances of latency, reliability and connection density as per ITU-R evaluation methodology. We found that the latency requirements can only be met using half-slot transmissions or HARQ-less full-slot transmission in contrary to the results submitted in the self-evaluation report of the DECT-2020 NR. Due to lack of details in the submissions; like the channel and interference modeling between device to device relaying, scheduling and control of relay propagation direction, we could not conclude whether the DECT-2020 NR can meet the minimum performance requirement of connection density of IMT-2020.

Index Terms—5G India Forum, DECT, NR, Latency, Reliability, Connection Density, Beamforming

I. INTRODUCTION

The International Telecommunication Union Radiocommunication sector (ITU-R) has set the requirements for International Mobile Telecommunications-2020 (IMT-2020 5G Standard). The standard is about to complete in next few months that is, by 2020 [1]. However, some parts of it have been finalized earlier. For example, the requirements for radio access technologies (RAT) were adopted in ITU-R report M.2410 in November 2017 [2]. Following this publication, the developers of radio access technologies such as third generation partnership project (3GPP) started developing 5G technologies meeting these requirements. 3GPP has developed radio access technologies new radio (NR), Long term evolution - machine type communication (LTE-M) and narrow band - Internet of things (NB-IoT) which together meet all the requirements [3]. Additionally, many other candidate technologies developed by China, Korea, the European telecommunications standards

Institute (ETSI)-digital enhanced cordless telecommunications (DECT) Forum, enhanced ultra high throughput (EUHT) have been submitted to ITU-R; claiming to have met the minimum technology requirements.

Candidate technologies submitted to ITU-R need to fulfil a set of minimum technical requirements specified in ITU-R report M.2410 [2]. The report M.2412 [4] provides the guideline for the evaluation process for self-evaluation as well as independent evaluation. As per the ITU-R process, all the candidate technologies need to be independently evaluated by independent evaluation groups (IEG).

In this work, 5G India forum (5GIF) independent evaluation group has evaluated ETSI-DECT Forum candidate technology DECT-2020 NR. We have carried out analytical approach in this process. In this work, we focused on latency and reliability requirements necessary for mission critical applications and connection density requirements for massive machine type communication applications (mMTC). Since, no detailed specifications of the DECT-2020 NR was available at the time of evaluation and write up, our assessment is limited by the description template and self evaluation report submitted to ITU-R. We found, the ITU-R submission was not sufficient to even conclude on the connection density requirement. Additionally, due to the dependency of DECT-2020 NR technology on mesh based topology for delivering reliable machine type data packets in urban macro like test environment, the assessment requires channel and interference modeling information between device to device relaying, scheduling and propagation direction of relaying to guarantee packet delivery within 10s. We could not determine what is the minimum number of nodes needed in the mesh topology for the network to be fully connected. This could be lower than the minimum requirements set for connection density which misleads the definition of minimum technical performance requirement of connection density. Also, we found that the latency requirements can only be met using half-slot transmissions or hybrid acknowledge

request (HARQ)-less full-slot transmission in contrary to what was reported in self evaluation report of the DECT-2020 NR.

Rest of the paper contents are arranged as follows. Section II provides a brief of candidate technology DECT-2020 NR and minimum requirements. Section III presents details evaluation and observation. Section IV, concludes the paper.

II. DECT-2020 NR CANDIDATE TECHNOLOGY

This section provides information and data of self evaluation by technical committee (TC) DECT Forum. Before presenting their reports, it is appropriate to briefly list the major requirements for IMT 2020 in respect of reliability and connection density.

One of the critical requirements of IMT 2020 is ultra-low latency. It is the one way time taken to successfully deliver an application layer packet/message from the radio protocol layer 2/3 service data unit (SDU) ingress point to the radio protocol layer 2/3 SDU egress point of the radio interface in either uplink (UL) or downlink (DL). So, the latency of $1ms$ or less is the requirement. In terms of reliability, block error rates (BLERs) should be as low as 10^{-9} [6]. It is to be noted that the URLLC is one of the enabling technologies in the fourth industrial revolution [7]. For mMTC, the main performance requirement for a such network service is to support a high connection density of up to 1-million devices per km^2 which is 10-times the maximum amount currently possible with 4G LTE.

DECT is the ETSI standard for short-range cordless communications. DECT is suited to voice, data and networking applications with a range up to 500 meters. DECT-2020 NR is a major new standardization activity with the intention of meeting some of requirements of IMT-2020. DECT operates in the 1880–1900MHz band and defines ten frequency channels from 1881.792MHz to 1897.344MHz with a band gap of 1728kHz. Specifically, legacy DECT system is known to use 1880-1900MHz, 1900-1980MHz, 2010-2025MHz and 2400-2483.5MHz. DECT is limited to these bands. For the DECT-NR RIT component, the subcarrier spacing used in the specification is based on 27KHz. The maximum bandwidth that can be achieved with 27KHz sub-carrier spacing (SCS) and 1024 fast Fourier transform (FFT) points is 27.648MHz. The base station is referred as fixed point (FP) and mobile devices as portable points (PP).

In the DECT-2020 NR submission [8], the link budget reports same coverage, same transmission bit rate and same signal to noise ratio (SNR) for control&data for both uplink (UL) and downlink (DL). Parameters like Tx&Rx antenna ports, user equipment (UE) speed and information on outdoor to indoor (O2I) could not be found in the current report. Perhaps, they may submit in the revised document. DECT legacy use channel is scalable in multiples of 1.728MHz whereas, 3GPP supports multiple of 10MHz. Also, conventional legacy DECT uses omni-directional antennas whereas IMT 2020 3GPP heavily relies on active antenna system (AAS) and beamforming. It could not be found in current available

specifications that how DECT-2020 deploy beamforming in active AAS systems.

III. EVALUATION AND OBSERVATIONS

This section provides the details of the evaluation and findings for mMTC and URLLC usage scenario. It is to be noted that all the test environments set by ITU-R are only for evaluation purposes and may not exactly replicate the actual deployments including the carrier frequency and assumed bandwidth.

A. Latency

1) **User Plane Latency:** As stated, the minimum requirement for user plane latency is $1ms$ for URLLC usage scenario. The user plane latency only considers the latency of packets while the UE is already in a connected state and is aggregation of delay due to components like UE processing delay, frame alignment time, transmission time interval (TTI) for data packet transmission, hybrid automatic repeat and request (HARQ) re-transmission and base station (BS) processing delay. Fig. 1 shows the different segment of URLLC timings.

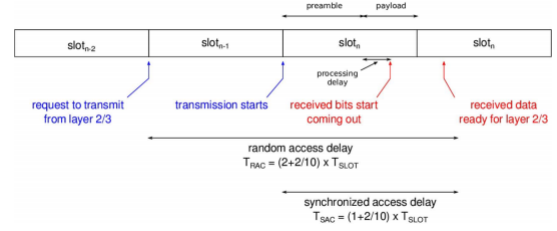


Fig. 1: URLLC timings

In DECT-2020 NR, low latency is achieved by configuring NR frame with alternating DL/UL slots. The low latency transmissions use a high-efficiency packet “full slot” (slot width is 0.416ms). Also, the aforesaid delay analysis comprised of: i) Transmitter processing delay of 1 OFDM symbol and receiver processing delay of 2 OFDM symbols; ii) In the case of HARQ transmission, maximum 6 slots for HARQ round-trip duration; iii) Frame alignment time (1 TTI) includes frame alignment within same current DL slot or extra waiting time for next available DL slot if the current slot is not DL slot. In the case of DECT UP latency, the calculation is identical for uplink and downlink cases.

TABLE I: Downlink User Plane Latency for 27KHz SCS NR TDD No HARQ (Frame Structure: DUDU)

Description	Half Slot(ms)	Full Slot(ms)
Avg symbol alignment time	0.0208	0.0208
BS pre-processing delay	0.0416	0.0416
Frame Alignment (max)	0.208	0.416
TTI for data packet transmission	0.208	0.416
UE pre-processing delay	0.0832	0.0832
Total one way UP latency	0.5616	0.9776

TABLE I shows that DECT-2020 NR achieves latency less than $1ms$ when no HARQ is used. TABLE II shows that when HARQ is used, the user plane latency value exceeds

TABLE II: Downlink UP Latency for 27KHz SCS NR TDD-HARQ (Frame Structure: DUDU)

Description	Half Slot(<i>ms</i>)	Full Slot(<i>ms</i>)
Avg symbol alignment time	0.0208	0.0208
BS pre-processing delay	0.0416	0.0416
Frame Alignment(max)	0.208	0.416
TTI for data packet transmission	0.208	0.416
UE pre-processing delay	0.0832	0.0832
Contribution by HARQ(10\$ BLER)	0.1248	0.2496
Total one way UP latency	0.6864	1.2272

1*ms* duration with full slot format. The technical study [7] provides the latency of 0.917*ms* and the value reported in the self evaluation report submission is 0.7904 *ms*. Also, in one of the of DECT Forum white paper, it is mentioned that the DECT-2020 NR technology achieves a latency between 2 and 10*ms* [9].

2) **Control Plane Latency:** The control plane (CP) latency refers to the transition time from a most “battery efficient” state (e.g. Idle state) to the start of continuous data transfer (e.g. Active state). This requirement is defined for the purpose of evaluation in the eMBB and URLLC usage scenarios. The minimum requirement for control plane latency is 20*ms*. The CP latency delay analysis comprises: i) Latency due to random access procedure; ii) UL synchronization; iii) Connection establishment and; iv) Data bearer establishment in addition to HARQ re-transmission. Fig. 2 shows CP procedure to support the above statements.

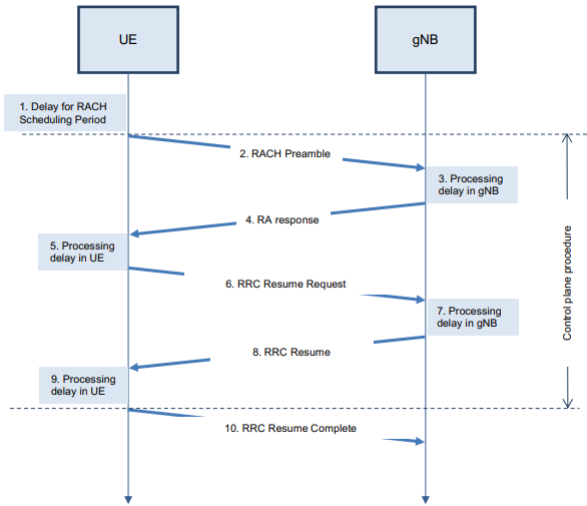


Fig. 2: CP procedure (NR Release 15)

TABLE-III provides signaling delay from non-associated to associated for 27KHz SCS. The mentions of ETSI about separate use of random access channel for better protection of critical traffic (e.g. URLLC) has been proposed as a working idea that would need to be confirmed in the MAC design phase [10]. Also, reference to the physical random access channel (PRACH) timings (*Xms* & *Yms*) could not be found in the current specification submitted by ETSI [10]. This might

TABLE III: Signalling delay from non-associated to associated for 27KHz SCS

Step	Description	Delay(<i>ms</i>)
1	Delay due to RACH scheduling period	0
2	Transmission of RACH preamble	X
3	Preamble detection and processing in gNB	Y
4	Transmission of RA response	0.4167
5	UE processing delay	5 <i>ms</i>
6	Transmission of RRC resume request	0.4167
7	Association request processing time	1 or 5
8	Association response TX response	0.4167
9	Association response processing time	1
	Total	>(8/12)

subject to possible inclusion of this information in updated version of their specifications. As shown in the TABLE-III, DECT-2020 NR meets the control plane latency requirements.

B. Reliability

Reliability is the success probability of transmitting a layer 2/3 packet within a required maximum time. The minimal requirement defined in report ITU-R M.2410 [2] is $1-10^{-5}$ success probability of transmitting a layer 2 PDU (protocol data unit) of 32bytes within 1*ms*. Here, reliability is evaluated under urban macro – URLLC test environment. The reliability evaluation uses system-level simulation followed by link-level simulation. Especially, the evaluation configuration B (carrier frequency = 700MHz) and channel model A are evaluated for DL and UL. The detailed evaluation assumptions for system-level and link-level simulations are provided in TABLE-IV and TABLE-V respectively.

Downlink Case

In DECT-2020 NR evaluation, frequency reuse schemes are exploited to mitigate interference and improve the reliability. The following three configurations for frequency reuse factor 1, 3, and 7 are evaluated [8].

- **Case 1:** The frequency reuse factor is set to 1. A single DECT-2020 NR channel with 1.728MHz bandwidth is applied for URLLC service in each cell, i.e. the system can be considered as a single frequency network.
- **Case 2 & Case 3:** The frequency reuse factor is set to 3 in the case 2 and 7 in the case 3. Accordingly, DECT-2020 channels are applied for URLLC service and the neighboring BSs use different channels.

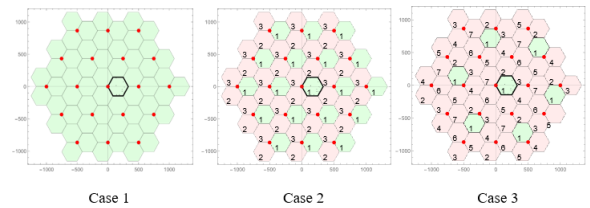


Fig. 3: Evaluated URLLC system configuration.

The network layouts for different frequency reuse factors are provided in Fig. 3. In this, the interference cell from the wrap-around layout are not marked. In the system-level simulation,

TABLE IV: DECT System Level Evaluation Assumptions for DL/UL Reliability

Configuration Parameters	URLLC Configuration B
Inter-site Distance	500 m
Base Station Antenna Height	25 m
UE Antenna Height	1.5m
Antenna Elements per TxRP	60 Elements(15x4)
UE Antenna Elements	4
Device Deployment	80% Outdoor, 20% Indoor
BS Noise Figure	5dB
UE Noise Figure	7dB
BS Antenna Element Gain	8dBi
UE Antenna Element Gain	0dBi
Thermal Noise Level	-174dBm/Hz
Traffic Model	Full Buffer
Simulation Bandwidth	20MHz
UE Density	10UEs per TxRP
Numerology	27KHz SCS
Scheduling	PF
Receiver	MMSE
Channel Estimation	Non-Ideal
Carrier Frequency	700MHz
TxRP/Site	3
Wrapping Around Method	Geographical Distance
Criteria for Evaluation of Serving TxRP	RSRP Based
Mechanical Tilt	90°
Electrical Tilt	99°
SLAV	30
HBeamwidth	65
VBeamwidth	65
Horizontal Scan	0.0
Horizontal Spacing b/w Antenna Elements	0.5
Vertical Spacing b/w Antenna Elements	0.8

the signal-to-interference-noise-ratio (SINR) distributions for different frequency reuse factors is shown in Fig. 4. and the fifth percentile (5%-tile) SINR are illustrated in TABLE VI. Pre-processing SINR is used for reliability evaluation which is defined on Rx antenna port with respect to a Tx antenna port.

The link-level simulation assumptions are available in TABLE V. The SNR-BLER curve is illustrated in Fig. 5. Based on the results from Fig. 4 and Fig. 5, the DL reliability is obtained as provided in TABLE-VI. Results are evaluated for 27KHz SCS, SU-MIMO (BS antenna array: 15x4) in Non Line of Sight (NLOS) channel condition. It is observed that DECT-2020 NR reliability is marginally lower than the ITU requirement in DL using the maximum antenna array 15x4.

TABLE V: DECT Link Level Evaluation Assumptions for DL/UL Reliability

Configuration Parameters	URLLC Configuration B
Evaluated Service Profiles	Full Buffer Best Effort lex_{lex}
Simulation Bandwidth	1.728 MHz
Number of users in simulation	1
Packet Size	37 Bytes at Layer 2 PDU
Link Level Channel Model	TDL-iii
Delay Spread Scaling Parameter	363 ns
Carrier Frequency	700 MHz
Numerology	27 KHz SCS
Number of Antenna Elements/TxRP	60 Elements(15x4)
UE Antennas	4
Packet Format	Long Preamble Packet
Channel Estimation	Non Ideal
Number of Symbol for Control Information	2
Number of Symbol for data	4
Control Information Modulation and Coding	TBCC with Code Rate=1/2, QPSK Repetition 2
Data Modulation and Coding	Turbo with Code Rate=3/4, QPSK

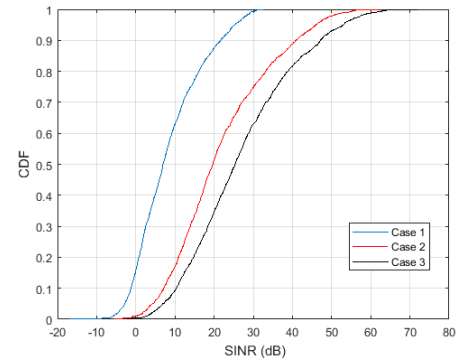


Fig. 4: Downlink SINR distribution obtained from system level simulation (BS antenna array: 15x4, BS Tx power: 49dBm)

Uplink Case

For uplink reliability evaluation, the frequency reuse schemes are the same as that of downlink. In the system-level simulation, the SINR distributions for different frequency reuse factors are provided in Fig. 6. and the fifth percentile (5%-tile) SINR is illustrated in TABLE VII.

Here also, we have used the same setting. The SNR-BLER curve is illustrated in Fig. 5. Based on the results from Fig. 5 and Fig. 6, the uplink reliability is obtained in TABLE VII. It is observed that the uplink reliability value is marginally lower than ITU-R requirement using the maximum antenna array size of 15x4.

We have also shown the antenna radiation pattern and gains at different locations. Fig. 7 and Fig. 8 shows the antenna gains

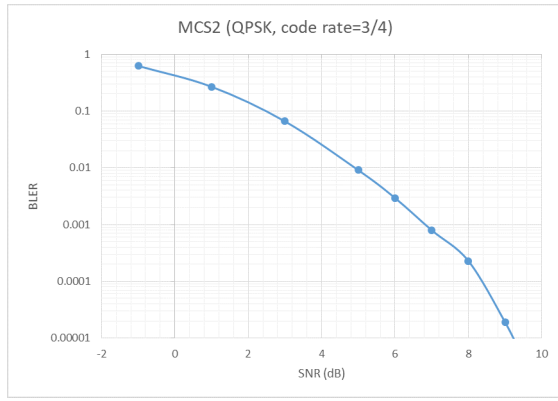


Fig. 5: SNR-BLER curve for data channel evaluation (BS antenna array: 15x4, BS Tx power: 49 dBm)

TABLE VI: 5%-tile SINR obtained from system-level simulation and evaluation results of downlink reliability

Configuration	Case1	Case2	Case3
5%-tile SINR (BS Tx power: 49 dBm BS antenna array: 15x4)	-2.8 dB	4.4 dB	6.9 dB
Reliability	10.9213%	98.3007%	99.9215%

available at different locations in the network layout when one BS is active (considering 3-transmission reception points (TRxP)) for DECT-2020 NR and 3GPP respectively. From the Fig. 7, it is observed for DECT-2020 NR that good antenna gains are obtained not only at locations near BS but the users at far location are also getting some gains which could add up to give high interference to that particular user from this active base station. In Fig. 8, the 3GPP shows good gains in the cell itself with good coverage for nearby users with negative gains in locations away from the base station.

Fig. 9 shows 2D antenna radiation pattern for 15x4 antenna configuration with electric tilt 99°, mechanical tilt 90°, escan 0°. The plot with labelling 'Azimuth' shows antenna gains along azimuth angle with elevation fixed at 99° and the plot labelled 'Elevation' shows antenna gains along elevation angle with azimuth angle fixed at 0°. It can be seen from figure that in case of both the plots, gain is maximum at point where azimuth angle is 0° and elevation angle is 99° in coordination with the electric tilt and horizontal scan provided for beamforming. Similar observation is noted from Fig. 10 which shows 3D antenna radiation pattern for the same. Also, it can be observed that as number of elements are more in vertical direction, beam is narrow in vertical and wider in horizontal direction.

C. Connection Density

Connection density is the total number of devices fulfilling a specific quality of service (QoS) per unit area (per km^2). The requirement is fulfilled if the 99th percentile of the delay per user is less than or equal to 10s and the connection density is

TABLE VII: 5%-tile SINR obtained from system-level simulation and evaluation results of uplink reliability

Configuration	Case1	Case2	Case3
5%-tile SINR (BS Tx power: 49 dBm BS antenna array: 15x4)	-0.4 dB	3.7 dB	4.1 dB
Reliability	48.5944%	96.3088%	97.4825%

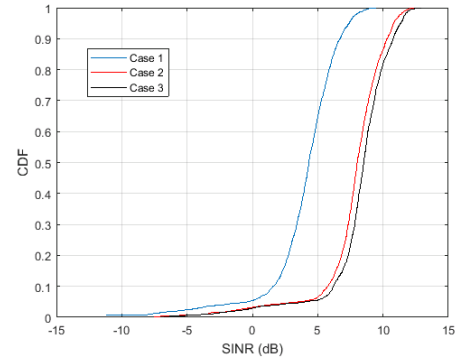


Fig. 6: Uplink SINR distribution obtained from system level simulation (BS antenna array: 15x4, UE Tx power: 23dBm)

greater than or equal to 1,000,000 devices per km^2 . Following observation were made while evaluating DECT-2020 NR's connection density:

- DECT Forum assumes a MESH based topology and relies on multi-hop communication to get the device from a MTC device to the network. The link budget for mMTC though claims to have a coverage of data, control channel of 480m with 100% reliability, but in the self-evaluation it is stated that DECT with star topology does not meet coverage requirement due to which multi-hop mesh technology is implemented in mMTC scenario.
- Since the connection density evaluation of DECT depends on "relaying", assuming the technology could meet the minimum requirement of connection density of 1,000,000 devices per km^2 , it does not ensure that if the number of devices deployed are very less then the packet data can be delivered from such devices in such macro cells due to lack of sufficient devices for relaying.
- The evaluation of relay-based simulation requires channel model between device to device which is at the same height (1.5m), interference modeling information between device to device relaying, scheduling and propagation direction of relaying to guarantee packet delivery. This is not supported in the current channel model in M.2412. Details about it could not be found in the report.
- Interference characteristics and modeling is also needed to understand the quality of multi-hop relaying to ensure the small packets are delivered to the final network within the given time with 1% PER probability. The probability of successful transmission (link reliability) reduces and probability of bit error increases as the density of inter-

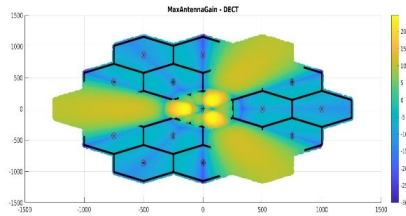


Fig. 7: Maximum Antenna Gain Possible in a Network Layout for DECT with one BS Active.

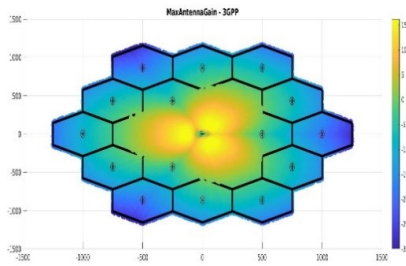


Fig. 8: Maximum Antenna Gain Possible in a Network Layout for 3GPP with one BS Active.

fering nodes in the interference region increases [11].

- It could not be found in the self-evaluation report that how the “relay propagation” from one mMTC device is restricted to flow to adjacent cells.

IV. CONCLUSION

This paper provided the evaluation results and observations on the candidate technology component DECT-2020 NR which was submitted by the ETSI DECT Forum. This component technology needs to meet the performance requirements for URLLC and mMTC usage scenario. It is observed that DECT-2020 NR meets the control plane latency requirement of $20ms$ and the user plane latency requirement of $1ms$ by enabling HARQ-less transmission. But the user plane latency exceeds the $1ms$ requirement when full slot transmission with HARQ is used. Based on parameters and limited details of specifications, we conclude that the assumed antenna parameters, inter-cell interference there are many UEs with low SINR values in DL as well as UL. Due to poor geometric SINR distribution, the performance requirement of reliability was found marginally lower than the minimum requirements for IMT-2020. The DECT 2020 NR has the worst SINR distribution with a single frequency reuse deployment. The final submissions from ETSI DECT Forum to ITU-R lacks detailed specifications regarding the usage of mesh topology which was necessary to evaluate the supported connection density; using the ITU-R evaluation framework. The DECT-2020 NR technology inherently depends on mesh topology for delivering machine type data packets in the urban macro test environment, it is ambiguous what will happen when only few IoT devices are deployed in a macro like scenario. Further, investigation is required to evaluate the minimum density of

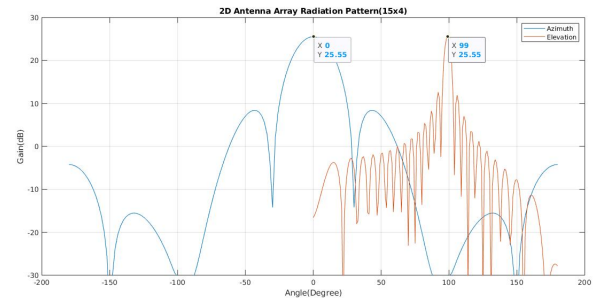


Fig. 9: 2D Antenna Radiation Pattern along Elevation and Azimuth(15x4)

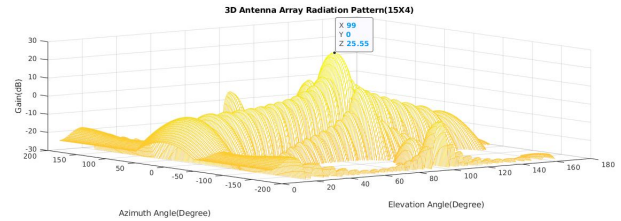


Fig. 10: 3D Antenna Radiation Pattern(15x4)

devices needed for reliable relaying of small packets. When DECT-2020 NR is deployed for smaller or indoor industrial environment, the dependency on multi-hop to reliably deliver the small machine-type packet data might be less stringent. These aspects could be assessed when detailed specifications of DECT 2020 NR are available.

V. ACKNOWLEDGEMENT

This work is supported by 5G India Forum (5GIF) Independent Evaluation Group (IEG) for IMT-2020: 5G India Forum is associated with COAI, India.

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